

INSTITUTO TECNOLÓGICO AUTÓNOMO DE MÉXICO

FUNDAMENTALS OF FINANCE

Professor: Juliana Gudiño Antillón

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Subject Code: ADM-15507

PRERREQUISITES:

ADM-12107 Strategy I

CON-10001 Introduction to Accounting

ECO-11101 Economics I

GENERAL OBJECTIVE:

Fundamentals of Finance is an introductory course, which aims to study the general Principles in Finance and give students a complete overview of Finance. The basis of specialized courses will be established in this course.

The course of Fundamentals of Finance covers all the Finance subthemes under the same conceptual framework.

It is expected that students acquire knowledge about the Principles of Corporate Finance, the Functioning of Financial Markets, Investments, and Modern Valuation Techniques for Fixed Income Securities, Equities and Derivatives.

GENERAL ASPECTS:

The study of Finance refers to the allocation of scarce resources over time under uncertainty conditions. Considering this approach, this course is oriented to provide students the knowledge of the three related pillars of finance that support financial theory, which are interrelated:

- a) Optimization over time (exchange relations over time)
- b) Valuation of financial securities (bond and stocks)
- c) Corporate Finance

In relation to the course content, the first part pretends that students understand the mechanism through which individuals select among different alternatives with uncertain payments, specifically refers to consumption and investment decisions. Also, in this part of the course, it is highlighted the importance of a basic financial precept “One dollar today values more than one dollar tomorrow”, traditionally known as Time Value of Money. This same concept will be used for valuing bonds and stocks. Additionally, the term structure of interest rates are going to be studied, because is one of the relevance issues of the Fixed Income Market.

In a second stage, the course will focus on the study of the different techniques used for valuing investment projects under conditions of certainty and uncertainty, analyzing the advantages and disadvantages of each one. There will be an important emphasis in the building of cash flows, and also in the calculation of the Weighted Average Cost of Capital.

In the third stage, the course will be focused on the study of the principles of Portfolio Investment Theory, in particular in the concepts of risk preference and risk aversion. The mean variance model will be studied, specially highlighting the relation between risk and return, which is a basic precept for any financial decision considering the benefits of diversification.

Finally, students will study the basic concepts about Derivatives. The main objective is that students will get related with the terminology used in Derivatives, and acquire the knowledge and understanding of the characteristics of such instruments.

METHODOLOGY:

This class will consist of teacher presentations, part of the material will be presented in slides and other in the blackboard. Students should bring printed slides to the class, to facilitate the understanding of class developments made in the blackboard.

During the semester the teacher will assign homework with a specific delivery date, the teacher won't receive any homework in other date than that initially established.

Students must solve during the semester some brief exams with duration of 10 minutes, in that way it is expected that students could measure their learning and course comprehension.

Professor will assign some readings during the semester, which must be commented by students during the next class or a report about the reading must be delivered as an assignment.

There will be two partial exams and a final cumulative exam, all them will cover 80% of the final grade. Each exam will be comprehensive with 2 hours duration. The exam will be integrated by different types of questions, multiple choice theoretical questions, open numerical exercises, proofs, multiple choice numerical exercises. All the exams will be applied in a computer room, and student could use a spreadsheet or a financial calculator to solve them.

Each week, there will be a lab class, of 1.30 hour duration, which will be taught by a student of the Financial Management degree whom passed this course with an excellent score. Each week, professor will attach in her webpage the lab concerned the theme studied. The objective of solving labs is that students could comprehend better the material covered in the classroom, and it is expected that they reinforce the learning process.

COURSE ORGANIZATION:

THEME	BOOK CHAPTERS	CLASSES / # HOURS
I. Introduction to Finance A. Goals and Governance of the Firm - Goals in Corporate Finance - Goals of the Corporation - The Role of the Financial Manager B. Financial System - Money and Capital Markets - Primary and Secondary Markets - Fixed Income Securities and Variable Income Securities C. Code of Ethics in Finance Code of Ethics and Standards of Professional Conduct http://www.cfapubs.org/doi/pdf/10.2469/ccb.v2010.n14.1 STANDARDS OF PRACTICE HANDBOOK: http://www.cfapubs.org/doi/pdf/10.2469/ccb.v2010.n2.1	1 y 2	2.5 classes 3.75 hours
II. Consumption and Investment Decisions A. Consumption and Investment Decisions B. Relationship between Consumption and Investment Decisions and Financial Markets C. The Fisher Separation Theorem D. Analysis of Investment Projects	Copeland Weston 1	2.5 classes 3.75 hours
III. Financial Mathematics A. Time Value of Money. Present Value and Future Value. B. Simple Interest Rate and Compound Interest Rate. Nominal Interest Rate, Effective Interest Rate and Equivalent Interest Rates. Nominal and Real Rates of Interest. C. Discount Interest Rates and Yield Rates. D. Annuities and Perpetuities with and without growing. E. Amortization Schedules - Constant Payments - Constant Principal Repaid - Adhoc Schemes	3	8.5 classes 12.75 hours
IV. Valuation Models A. Principles of Assets Valuation B. Valuing Bonds - Basic Concepts and Classification - Bond Pricing - Yield Measures - Duration C. Term Structure of Interest Rates D. The Value of Common Stocks - Basic Concepts and Classification	4 y 5	7 classes 10.5 hours

2. Stock Pricing and Multiples		
V. Valuation of Investment Projects A. Introduction to Valuation of Investment Projects 1. The Free Cash Flow 2. Weighted Average Cost of Capital (WACC) 2.1 Determinants of Debt Cost 2.2 Determinants of Capital Cost 3. Capital Structure and Leverage B. Investment Criteria 1. Payback Period 2. Discounted Payback Period 3. Net Present Value 4. Internal rate of Return (IRR) 5. Profitability Index C. Project Analysis under Uncertainty 1. Different Life Projects 2. Optimum Time for Replacement 3. Decision Trees 4. Sensitivity Analysis, Scenario Analysis and Beak-Even point	6, 7, 10 y 11	7 classes 10.5 hours
VI. Valuing Assets A. Portfolio Theory B. Capital Asset Pricing Model (CAPM) C. Market Efficiency D. Introduction to Derivatives 1. Futures 2. Options	8, 9, 14 y 21	4.5 classes 6.75 hours

GRADING:

First Exam	20%
Second Exam	20%
Final Exam (cumulative)	40%
Assignments and Quizzes:	20%
Assignments	10%
Quizzes	10%

Notes:

- The final exam will be cumulative
- Partial exams will be on Friday at 2:00 p.m.

COURSE POLICIES:

- Professor won't receive assignments in other date than that established
- The final grade will be rounded to the following digit only if the final average is 6.5 -9.5
- Students must obtain a minimum final average of 6.0 for passing the course.
- If it appears that some student has copied in an examination, homework or quiz, shall be considered a grade of zero on it and proceed in accordance to ITAM's regulation.
- It is necessary that the final exam be approved for passing the course
- There are only 5 minutes tolerance to enter the class
- Cell phones and similar must be turned off during the class
- If a student does not present some of the exams will be assigned a grade of zero on it, except for important cases clearly justified, in which case the corresponding share of the examination will be taken on the final exam.

SUGGESTED CALCULATORS (OPTIONAL):

Texas Instrument BAII Plus; HP 12 C ; HP 12 C Platinum

SUGGESTED SOFTWARE (OPTIONAL):

Matlab; Maple; Excel

BASIC REFERENCES:

Brealey, R. Myers, S, and Allen, F., 2008, Principles of Corporate Finance. Ninth Edition, New York, USA: McGraw Hill-Irwin.

COMPLEMENTARY REFERENCES:

Copeland E. Thomas and Weston J. Fred, 2005, Financial Theory and Corporate Policy., Addison Wesley, Fourth Edition.

Bodie Zvi, Merton C. Robert., 2008, Finance. New Jersey, USA, Prentice Hall.

Elton J. Edwin y Gruber J. Martin., 2007, Modern Portfolio Theory and Investment Analysis. USA. John Wiley and Sons, Inc., Seventh Edition.

Fabozzi, Frank, 1996, Bond Markets Analysis and Strategies, Prentice Hall.

Hawawini G. and Viallet Claude, 2002, Finance for Executives. Managing for Value Creation, South Western. Second Edition. USA.

Haugen, R., 1993, Modern Investment Theory, Third Edition, Englewoods Cliffs, Prentice Hall.

Hull, J., 2009, Options, Futures and Other Derivatives, Seventh Edition, Prentice Hall, Englewood Cliffs, N.J.

Kellison G. Stephen, 1991, The Theory of Interest, Second Edition, USA. Irwin: McGraw-Hill.

Ross, S., Westerfield J., Jaffe J., 2010, Corporate Finance. Ninth Edition. New York, USA: McGraw Hill-Irwin.

TECHNICAL NOTES (Mandatory):

Gudiño, Juliana. (2008). *Tasas de interés: nominales, efectivas y equivalentes*. Nota Técnica. Serie de Casos de la División Académica de Administración y Contaduría. Clave 082-08-FIN-NT. México: ITAM.

Gudiño, Juliana. (2008). *Anualidades y Perpetuidades*. Nota Técnica. Serie de Casos de la División Académica de Administración y Contaduría. Clave 083-08-FIN-NT. México: ITAM.

Gudiño, Juliana. (2008). *Tablas de Amortización*. Nota Técnica. Serie de Casos de la División Académica de Administración y Contaduría. Clave 085-08-FIN-NT. México: ITAM.

COURSE PRINCIPLE:

Students are enjoined to discuss with their peers the school material covered in the course, however, assignments and tests must be individual work of each student or team as appropriate.

Any violation of the rules of ITAM's regulation will be resolved in strict compliance.